

- As with any antipsychotic, use with caution in patients with history of seizures
- Antipsychotics may cause somnolence, postural hypotension, motor and sensory instability, which may lead to falls; for patients with diseases, conditions, or medications that could exacerbate these effects, complete fall risk assessments when initiating antipsychotic treatment, and recurrently for patients on long-term antipsychotic therapy

Do Not Use

- If there is a proven allergy to brexpiprazole

Special Populations

Renal Impairment

- Moderate, severe, or end-stage: maximum recommended dose for depression is 2 mg once daily and for schizophrenia is 3 mg once daily

Hepatic Impairment

- Moderate to severe: maximum recommended dose for depression is 2 mg once daily and for schizophrenia is 3 mg once daily

Cardiac Impairment

- Use in patients with cardiac impairment has not been studied, so use with caution because of risk of orthostatic hypotension
- Use with caution if patient is taking concomitant antihypertensive or alpha 1 antagonist

Elderly

- Some elderly patients may tolerate lower doses better
- Although atypical antipsychotics are commonly used for behavioral disturbances in dementia, no agent has been approved for treatment of elderly patients with behavioral symptoms of dementia such as agitation
- Elderly patients with dementia-related psychosis treated with atypical antipsychotics are at an increased risk of death compared to placebo, and also have an increased risk of cerebrovascular events
- Consider pimavanserin for dementia-related psychosis or Parkinson's disease psychosis instead of an atypical antipsychotic



Children and Adolescents

- Safety and efficacy have not been established
- Children and adolescents using brexpiprazole may need to be monitored more often than adults and may tolerate lower doses better



Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- In animal studies, brexpiprazole did not demonstrate teratogenicity
- There is a risk of abnormal muscle movements and withdrawal symptoms in newborns whose mothers took an antipsychotic during the third trimester; symptoms may include agitation, abnormally increased or decreased muscle tone, tremor, sleepiness, severe difficulty breathing, and difficulty feeding
- Psychotic symptoms may worsen during pregnancy and some form of treatment may be necessary
- Brexpiprazole may be preferable to anticonvulsant mood stabilizers if treatment is required during pregnancy
- National Pregnancy Registry for Atypical Antipsychotics: 1-866-961-2388,
<http://womensmentalhealth.org/research/pregnancyregistry/atypicalantipsychotic/>

Breast Feeding

- Unknown if brexpiprazole is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- * Recommended either to discontinue drug or bottle feed unless the potential benefit to the mother justifies the potential risk to the child
- Infants of women who choose to breast feed while on brexpiprazole should be monitored for possible adverse effects

The Art of Psychopharmacology

Potential Advantages

- For patients who do not tolerate aripiprazole or cariprazine

Potential Disadvantages

- Expensive

Primary Target Symptoms

- Positive symptoms of psychosis
- Negative symptoms of psychosis
- Cognitive symptoms
- Unstable mood (both depression and mania)
- Aggressive symptoms

- Approved as an adjunct treatment for depression
- Animal data suggest that brexpiprazole may improve cognitive impairment in schizophrenia
- Brexpiprazole is in late-stage clinical testing for agitation associated with Alzheimer dementia
- Brexpiprazole has shown evidence of efficacy especially in combination with sertraline for posttraumatic stress disorder
- Not approved for mania, but almost all atypical antipsychotics approved for acute treatment of schizophrenia have proven effective in the acute treatment of mania as well
- Pharmacological differences from aripiprazole suggest less akathisia with brexpiprazole, but no head-to-head trials
- Compared to aripiprazole, brexpiprazole has more potent binding of several receptor sites relative to dopamine 2 receptor binding, namely 5HT1A, 5HT2A, and alpha 1 receptors; however, the clinical significance of these differences is still under investigation

The Art of Switching



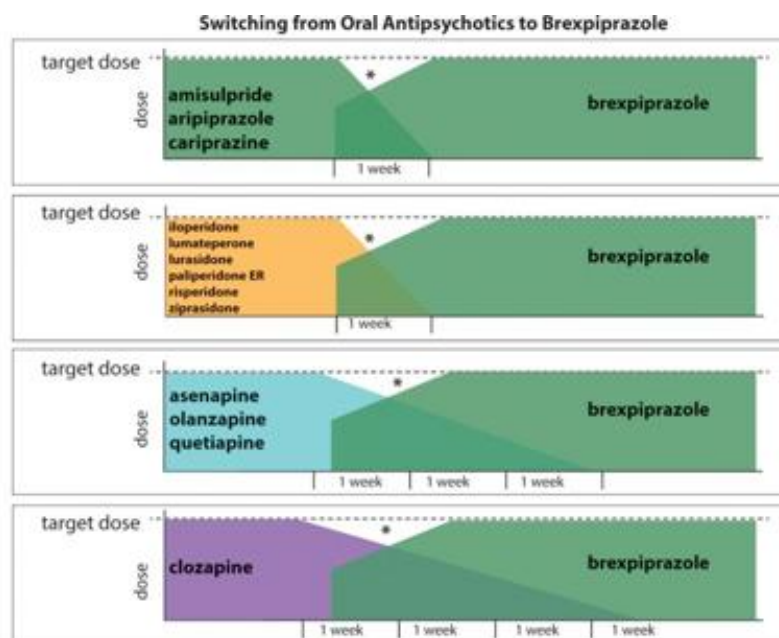
Switching from Oral Antipsychotics to Brexpiprazole

- It is advisable to begin brexpiprazole at an intermediate dose and build the dose rapidly over 3–7 days
- Clinical experience has shown that asenapine, quetiapine, and olanzapine should be tapered off slowly over a period of 3–4 weeks,

to allow patients to readapt to the withdrawal of blocking cholinergic, histaminergic, and alpha 1 receptors

- Clozapine should always be tapered off slowly, over a period of 4 weeks or more

* Benzodiazepine or anticholinergic medication can be administered during cross-titration to help alleviate side effects such as insomnia, agitation, and/or psychosis



Suggested Reading

Huhn M , Nikolakopoulou A , Schneider-Thoma J , et al. Comparative efficacy and tolerability of 32 oral antipsychotics for the acute treatment of adults with multi-episode schizophrenia: a systematic review and network meta-analysis . *Lancet* 2019 ;**394** :939–51 .

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Kane JM , Skuban A , Ouyang J , et al. A multicenter, randomized, double-blind, controlled phase 3 trial of fixed-dose brexpiprazole for the treatment of adults with acute schizophrenia . *Schizophr Res* 2015 ;**164** (1 –3):127–35 .

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Kishi T , Sakuma K , Nomura I , et al. Brexpiprazole as adjunctive treatment for major depressive disorder following treatment failure with at least one antidepressant in the current episode: a systematic review and meta-analysis . *Int J Neuropsychopharmacol* 2019 ;**22** (11):698 –709 .

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Ward K , Citrome L . Brexpiprazole for the maintenance treatment of adults with schizophrenia: an evidence-based review and place in therapy . *Neuropsychiatr Dis Treat* 2019 ;**15** :247–57 .

Buprenorphine

Therapeutics

Brands

- Suboxone (with naloxone)
- Probuphine (implant)
- Sublocade (subcutaneous injection)

see index for additional brand names

Generic?

Yes (not for implant or injection)

Class

- mu opioid receptor partial agonist

Commonly Prescribed for

(bold for FDA approved)

- **Induction of treatment for opioid dependence (Bunavail only)**
- **Maintenance treatment of opioid dependence (sublingual)**
- **Maintenance treatment of opioid dependence in patients who have achieved and sustained prolonged clinical stability on low-**

to-moderate doses (no more than 8 mg) of a transmucosal buprenorphine-containing product (implant)

- **Moderate to severe opioid use disorder in patients who have initiated treatment with a transmucosal buprenorphine-containing product, followed by dose adjustment for a minimum of 7 days (injection)**

How the Drug Works

- Binds with strong affinity to the mu opioid receptor, preventing exogenous opioids from binding there and thus preventing the pleasurable effects of opioid consumption
- Because buprenorphine is a partial agonist, it can cause immediate withdrawal in a patient currently taking opioids (i.e., reduces receptor stimulation in the presence of a full agonist) but can relieve withdrawal if a patient is already experiencing it (i.e., increases receptor stimulation in the absence of a full agonist)
- Buprenorphine is also an antagonist at the kappa opioid receptor
- In combination with naloxone: naloxone is a mu opioid receptor antagonist and can therefore block the effects of buprenorphine; however, because naloxone has poor sublingual bioavailability, it does not interfere with buprenorphine's effects when used properly. Naloxone does have good parenteral bioavailability; thus, if one tries to administer the buprenorphine/naloxone formulation intravenously, naloxone will prevent any rewarding effects from buprenorphine

How Long Until It Works

- Effects on withdrawal can be immediate
- Effects on reducing opioid use disorder/dependence can take many months of treatment

If It Works

- Reduces cravings, decreases opioid consumption
- Reduces effects of opioid withdrawal
- Diminishes rewarding effects of opioid consumption

If It Doesn't Work

- Evaluate for and address contributing factors
- Consider switching to another agent
- If patients receiving the implant feel a need for supplemental dosing, they should be evaluated and transmucosal buprenorphine should be considered



Best Augmenting Combos for Partial Response or Treatment Resistance

- Augmentation with behavioral, educational, and/or supportive therapy in groups or as an individual is probably key to successful treatment
- Buprenorphine can be prescribed in combination with naloxone (Suboxone) to decrease the potential for abuse or diversion

Tests

- Liver function tests at baseline and during treatment

Side Effects

How Drug Causes Side Effects

- Binding at mu opioid receptors

Notable Side Effects

- Headache, constipation, nausea
- Oral hypoesthesia, glossodynia
- Orthostatic hypotension
- Implant specific: insertion site pain, pruritis, erythema



Life-Threatening or Dangerous Side Effects

- Respiratory depression
- Hepatotoxicity

Weight Gain



- Reported but not expected

Sedation



- Many experience and/or can be significant in amount

What to Do About Side Effects

- Wait
- Reduce dose
- Switch to another agent

Best Augmenting Agents for Side Effects

- Dose reduction or switching to another agent may be more effective since most side effects cannot be improved with an augmenting agent

Dosing and Use

Usual Dosage Range

- Sublingual: 8–32 mg/day
- Implant: 4 implants intended to be in place for 6 months
- Injection: maintenance dose is 100–300 mg monthly

Dosage Forms

- Sublingual tablet 2 mg, 8 mg
- Sublingual tablet (with naloxone) 2 mg/ 0.5 mg, 8 mg/ 2 mg

- Sublingual film (with naloxone) 2 mg/0.5 mg, 4 mg/1 mg, 8 mg/2 mg, 12 mg/3 mg
- Each implant is 26 mm in length and 2.5 mm diameter and contains 74.2 mg of buprenorphine (equivalent to 80 mg of buprenorphine hydrochloride)
- Injection 100 mg/0.5 mL, 300 mg/1.5 mL

How to Dose –Sublingual

- Patients must be in a mild withdrawal state prior to starting bupre

Initiation (7 days)			
	Day 1	Day 2	Days 3–7
Buprenorphine	8 mg	12 or 16 mg	Increase in increments of 4 mg; maximum 32 mg
Buprenorphine/naloxone	8 mg/2 mg	12 mg/3 mg or 16 mg/4 mg	Increase in increments of 4 mg/1 mg; maximum 32 mg/8 mg

norphine

- Observe patient for at least 2 hours with initial dose, then have 1–2 visits in first week
- Achieve the lowest dose that eliminates withdrawal symptoms and illicit opioid use
- Stabilization (up to 2 months) and maintenance dose is generally 8–24 mg (8 mg/2 mg up to 24 mg/6 mg for buprenorphine/naloxone)
- During stabilization patients should be seen once per week
- During maintenance patients should be seen biweekly or monthly



Dosing Tips –Sublingual

- Buprenorphine must be administered sublingually, as swallowing reduces its bioavailability
- Patients should be instructed to place the sublingual formulation under the tongue and allow it to dissolve completely; the formulation should not be divided, crushed, chewed, or swallowed
- Can be dosed less often than once daily; one should double the dose for each additional 24-hour interval
- Buprenorphine alone is often used to initiate treatment, while buprenorphine/naloxone is preferred for stabilization and maintenance treatment
- Only buprenorphine with naloxone should be used for unsupervised administration, unless the patient has a proven allergy to naloxone

- Can be distributed through clinicians' offices by those who obtain a DEA DATA 2000 waiver
- Patients being switched between the 2 sublingual formulations (tablet and film) should be started on the same dose as the previously administered product; however, because the sublingual film has greater bioavailability than the sublingual tablet, patients must be monitored for over-medication (when switching from tablet to film) or under-medication (when switching from film to tablet); dose adjustment may be necessary

How to Dose –Implant

- Patient must have achieved and sustained prolonged clinical stability on transmucosal buprenorphine
- Four implants are inserted subdermally in the inner side of the upper arm for 6 months of treatment and are removed by the end of the sixth month



Dosing Tips –Implant

- Implants must be inserted and removed by trained healthcare providers who are certified in the Probuphine REMS program; information is available at www.probuphinerems.com or 1-844-859-6341
- Patients must currently be on a maintenance dose of 8 mg/day or less of transmucosal buprenorphine and should not be transitioned to a lower dose for the sole purpose of transitioning to the implant

- Patients should be on a stable transmucosal buprenorphine dose (8 mg/day or less) for 3 months or longer without any need for supplemental dosing or adjustments
- Examine the insertion site 1 week after implant insertion for signs of infection or other problems
- Patients should not receive prescriptions for transmucosal buprenorphine for as-needed use; patients who feel a need for supplemental dosing should be evaluated and alternative treatment should be considered
- For continued treatment, new implants may be inserted subdermally in an area of the inner side of either upper arm that has not been previously used at the time of removal
- If new implants are not inserted on the same day as removal, then patients should be maintained on their previous dosage of transmucosal buprenorphine
- After one insertion in each arm, most patients should be transitioned back to transmucosal buprenorphine if continued treatment is desired, as there is no experience with re-insertion into previously used administration sites or insertion into sites other than the upper arm

How to Dose –Injection

- Two monthly initial doses of 300 mg, followed by 100 mg monthly maintenance dose; can increase monthly maintenance dose to 300 mg if benefits outweigh risks

- Should be administered subcutaneously in the abdominal region; should not be administered intravenously or intramuscularly

Dosing Tips –Injection

- There should be a minimum of 26 days between doses
- Occasional delays in dosing of up to 2 weeks are not expected to have clinically significant impact on treatment effects

Overdose

- Can be fatal (less common than with methadone); respiratory depression, sedation, constricted pupils, bradycardia, hypotension, coma

Long-Term Use

- Maintenance treatment may be required; typical maintenance period is up to 2 years but may need to be indefinite

Habit Forming

- Buprenorphine is a Schedule III drug
- Can cause physical dependence

How to Stop

- Patients may experience a mild withdrawal syndrome if buprenorphine is stopped abruptly

- Taper to avoid withdrawal effects

Pharmacokinetics

- Metabolized by CYP450 3A4
- Elimination half-life of sublingual buprenorphine is 24–42 hours
- Elimination half-life of naloxone is 2–12 hours
- Implant: Tmax is 12 hours; time to steady state is 4 weeks

Drug Interactions

- Increased depressive effects, particularly respiratory depression, have occurred when taken with benzodiazepines or other CNS depressants; prescribe the lowest effective dose and shortest duration if taken concomitantly
- Plasma concentrations of buprenorphine may be increased by drugs that inhibit CYP450 3A4, so buprenorphine dose may need to be reduced if coadministered
- Patients taking a CYP450 3A4 inhibitor who transfer to the implant should be monitored to ensure that plasma buprenorphine levels are adequate
- If a CYP450 3A4 inhibitor is initiated in a patient with the implant, the patient should be monitored for signs of over-medication
- If a CYP450 3A4 inhibitor is discontinued in a patient with the implant, the patient should be monitored for signs of withdrawal

- Plasma concentrations of buprenorphine may be reduced by drugs that induce CYP450 3A4, so buprenorphine dose may need to be increased if coadministered
- Patients taking a CYP450 3A4 inducer who transfer to the implant should be monitored to ensure that plasma buprenorphine levels are not excessive
- If a CYP450 3A4 inducer is initiated in a patient with the implant, the patient should be monitored for signs of withdrawal
- If a CYP450 3A4 inducer is discontinued in a patient with the implant, the patient should be monitored for signs of over-medication

Other Warnings/Precautions

- Increased risk of CNS depressant effects when benzodiazepines and opioid medications are used together, including specifically the risk of slowed or difficult breathing and death
- If alternatives to the combined use of benzodiazepines and opioids are not available, clinicians should limit the dosage and duration of each drug to the minimum possible while still achieving therapeutic efficacy
- Although the risk is lower, buprenorphine can be abused in a manner similar to other opioids
- Parenteral misuse of buprenorphine/naloxone may result in marked opioid withdrawal syndrome

- To prevent withdrawal in patients dependent on opioids, patients must be in a mild withdrawal state prior to initiating treatment
- Attempts by patients to overcome blockade of opioid receptors by taking large amounts of exogenous opioids could lead to opioid intoxication or even fatal overdose
- Use with caution in patients with compromised respiratory function
- Risk of respiratory depression is increased with concomitant use of CNS depressants, particularly with parenteral administration
- Can cause severe, possibly fatal respiratory depression in children who are accidentally exposed to it
- Withdrawal symptoms can occur when switching from methadone to buprenorphine
- Buprenorphine may elevate cerebrospinal fluid pressure and should be used with caution in patients with head injury, intracranial lesions, and other circumstances when cerebrospinal pressure may be increased
- Buprenorphine may increase intracholedochal pressure and should be administered with caution to patients with dysfunction of the biliary tract
- Use with caution in debilitated patients and those with myxedema or hypothyroidism, adrenal cortical insufficiency (e.g., Addison's disease); CNS depression or coma; toxic psychoses; prostatic hypertrophy or urethral stricture; acute alcoholism; delirium tremens; or kyphoscoliosis

- Use the implant with caution in patients with a history of keloid formation, connective tissue disease, or history of recurrent recurrent methicillin-resistant *Staphylococcus aureus* (MRSA) infections
- Rare nerve damage and migration resulting in embolism and death may occur due to improper insertion of the implant in the upper arm; local migration, protrusion, and expulsion can also occur as a result of improper or incomplete insertion; protrusion and expulsion may occur as a result of infection
- In the event that an implant is expelled, the patient should store it in a plastic bag out of reach of children and bring it to their healthcare provider to ensure that the entire implant was expelled
- The prescribing healthcare provider will need to monitor the patient until the implant is replaced

Do Not Use

- As an analgesic
- If the patient is naive to opioid use
- If there is a proven allergy to buprenorphine
- If patient has severe hepatic impairment (buprenorphine/naloxone combinations only)
- If there is a proven allergy to naloxone (buprenorphine/naloxone combinations only)

Special Populations

Renal Impairment

- Dose adjustment not necessary

Hepatic Impairment

- In patients with moderate to severe impairment, plasma levels of buprenorphine can be higher and half-life can be longer; thus, these patients should be monitored for signs and symptoms of toxicity or overdose
- For severe impairment, the dose should be reduced
- Because dose adjustment is not possible with the implant, it is not recommended for use in patients with moderate to severe hepatic impairment
- Hepatic impairment results in reduced clearance of naloxone, so patients with severe impairment should not take buprenorphine/naloxone combinations; caution is warranted for patients with moderate impairment

Cardiac Impairment

- Use with caution

Elderly

- Use with caution

- Some patients may tolerate lower doses better

Children and Adolescents

- Safety and efficacy have not been established

Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- Buprenorphine may be preferable to methadone in pregnant women
- Neonatal withdrawal has been reported following use of buprenorphine during pregnancy
- In animal studies, adverse events have been observed at clinically relevant doses; no clear teratogenic effects were seen, but increases in skeletal abnormalities were observed in rats and rabbits given daily buprenorphine at doses 5.4 and 10.8 times the maximum human recommended dose
- Not generally recommended for use during pregnancy, especially during first trimester

Breast Feeding

- Some drug is found in mother's breast milk
- * Recommended either to discontinue drug or bottle feed

The Art of Psychopharmacology

Potential Advantages

- Patients with mild to moderate physical dependence

Potential Disadvantages

- Patients unable to tolerate mild withdrawal symptoms

Primary Target Symptoms

- Opioid dependence

Pearls

- Considered a “take home” medication that generally has less stigma and better adherence than methadone
- Relatively convenient to administer, with flexible dosing and ease of discontinuation

Suggested Reading

Bonhomme J , Shim RS , Gooden R , Tyus D , Rust G. Opioid addiction and abuse in primary care practice: a comparison of methadone and buprenorphine as treatment options . *J Natl Med Assoc* 2012 ;**104** (7–8):342–50 .

Jones HE , Finnegan LP , Kaltenbach K. Methadone and buprenorphine for the management of opioid dependence in pregnancy . *Drugs* 2012 ;**72** (6):747–57 .

Kraus ML , Alford DP , Kotz MM , et al. Statement of the American Society of Addiction Medicine Consensus Panel on the use of buprenorphine in office-based treatment of opioid addiction . *J Addict Med* 2011 ;**5** (4):254–63 .

Yokell MA , Zaller ND , Green TC , Rich JD. Buprenorphine and buprenorphine/naloxone diversion, misuse, and illicit use: an international review . *Curr Drug Abuse Rev* 2011 ;**4** (1):28 –41 .